

Python Assignment 1: Data Structures - Strings & Tuples

Overview

This project is a Python fundamentals assignment focused on **Strings and Tuples**.

The assignment demonstrates basic Python operations such as:

- String concatenation
- String indexing
- String slicing
- String reversal
- String methods
- Tuple creation
- Tuple concatenation
- Tuple repetition
- Tuple indexing and slicing

The main objective is to understand how strings and tuples are created, accessed, modified through supported operations, and processed using Python's built-in methods.

Strings (Concatenation, Slicing and Other methods) :

1. String Concatenation:

Write a Python program that takes two strings i.e string 1 “Hello ”, string 2 get name as input from the user and concatenates them together. Display the concatenated string as the output. Then concatenate string 3 “, welcome to Python programming” to the existing string and display the output string.

The program takes the user's name as input and combines it with the string:

"Hello "

The program then adds:

", welcome to Python programming"

Concepts Used

- `input( )`
- String variables
- String concatenation
- `+` operator

2. String Slicing and Indexing:

Write a Python program using the above concatenated string as input and performs the following tasks:

- Print the first character of the string.
- Print the last character of the string.
- Print the first 5 characters of the string.
- Print the last 11 characters of the string.
- Print the string in reverse.
- Use slicing and print the word “Python” from the existing string.

The concatenated string is used to perform different indexing and slicing operations.

Operations Performed

a. First Character

```
text[0]
```

Returns the first character of the string.

b. Last Character

```
text[-1]
```

Returns the last character.

c. First Five Characters

```
text[:5]
```

Returns the first five characters.

d. Last Eleven Characters

```
text[-11:]
```

Returns the last eleven characters.

e. Reverse the String

```
text[::-1]
```

Returns the string in reverse order.

f. Extract "Python"

The program locates the word “Python” and extracts it using slicing.

```
python_start = text.index("Python")
```

```
print(text[python_start:python_start + 6])
```

Concepts Used

- Positive indexing
- Negative indexing
- String slicing
- Reverse slicing
- `index()` method

3. String Methods:

Write a Python program that takes a string, `strM = "Python beginner tutorial"` and perform the following tasks:

- a. Convert the sentence to uppercase.
- b. Convert the sentence to lowercase.
- c. Use Capitalize and return the sentence to the original input form.
- d. Count the total number of occurrences of character 't' in the string.
- e. Replace all occurrences of "Python" with "Machine Learning" in the input string

strM = "Python beginner tutorial"

The following string is used:

strM = "Python beginner tutorial"

The program performs several string operations.

a. Uppercase

strM.upper()

Output:

PYTHON BEGINNER TUTORIAL

b. Lowercase

strM.lower()

Output:

python beginner tutorial

c. Capitalize

strM.capitalize()

Output:

Python beginner tutorial

d. Count Character 't'

```
strM.count("t")
```

This counts the occurrences of the character 't'.

e. Replace "Python"

```
strM.replace("Python", "Machine Learning")
```

Output:

Machine Learning beginner tutorial

Concepts Used

- upper()
- lower()
- capitalize()
- count()
- replace()

4.Tuples (Creation, Modification and Access) :

Create 1st tuple with values - (10, 20, 30), 2nd tuple with values - (40, 50, 60):

- a. Concatenate the two tuples and store it in "t_combine"
- b. Repeat the elements of "t_combine" 3 times
- c. Access the 3rd element from "t_combine"
- d. Access the first three elements from "t_combine"
- e. Access the last three elements from "t_combine"

Two tuples are created:

```
tuple1 = (10, 20, 30)
```

```
tuple2 = (40, 50, 60)
```

a. Concatenate Tuples

```
t_combine = tuple1 + tuple2
```

Result:

```
(10, 20, 30, 40, 50, 60)
```

b. Repeat Elements Three Times

```
t_repeat = t_combine * 3
```

Result:

```
(10, 20, 30, 40, 50, 60,  
 10, 20, 30, 40, 50, 60,  
 10, 20, 30, 40, 50, 60)
```

c. Access the Third Element

```
t_combine[2]
```

Result:

```
30
```

Python indexing starts from 0, so:

Index 0 → 10

Index 1 → 20

Index 2 → 30

d. Access First Three Elements

```
t_combine[0:3]
```

Result:

```
(10, 20, 30)
```

e. Access Last Three Elements

```
t_combine[-4:-1]
```

Result:

(40, 50, 60)

Expected Output

===== 1. STRING CONCATENATION =====

Enter your Name: Thara

Hello Thara

Hello Thara, welcome to Python programming

===== 2. STRING SLICING AND INDEXING =====

a. First character: H

b. Last character: g

c. First 5 characters: Hello

e. Reverse: ...

f. Python: Python

===== 3. STRING METHODS =====

a. Uppercase: PYTHON BEGINNER TUTORIAL

b. Lowercase: python beginner tutorial

c. Capitalize: Python beginner tutorial

d. Number of 't': 2

e. Replaced string: Machine Learning beginner tutorial

===== 4. TUPLES =====

a. Combined tuple: (10, 20, 30, 40, 50, 60)

b. Repeated tuple: (10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)

c. 3rd element: 30

d. First three elements: (10, 20, 30)

e. Last three elements: (40, 50, 60)

Conclusion

This assignment provided practical experience with fundamental Python data structures, particularly **Strings and Tuples**. The exercises demonstrated how to manipulate strings using concatenation, indexing, slicing, and built-in methods. Tuple operations such as concatenation, repetition, indexing, and slicing were also implemented.

The assignment helped build a strong foundation in Python programming and basic data structure operations.

THANKYOU